

**KIRIRI WOMEN'S UNIVERSITY OF SCIENCE AND TECHNOLOGY**

# INDUSTRIAL ATTACHMENT PLACEMENT PORTAL

## RESEARCH PROJECT

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**DIT/075/25**

**A RESEARCH PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE ACADEMIC REQUIREMENTS FOR THE AWARD OF DIPLOMA IN INFORMATION TECHNOLOGY OF KIRIRI WOMEN'S UNIVERSITY OF SCIENCE AND TECHNOLOGY.**

**APRIL 2026**

## DECLARATION

I hereby declare that this project is my original work and has not been submitted to any other institution or for any other award of Diploma in Information Communication Technology at Kiriri women's university of science and technology.

All sources of information used have been duly acknowledged,

**Student Name: …….**

**Registration number: ……**

**Date: ……...** **Signature: ……...**

## SUPERVISOR'S DECLARATION

I hereby declare that this project has been carried out under my supervision and guidance. To the best of my knowledge, it is the original work of the student and has not been submitted for any other academic award.

**Supervisor's name: …….**

**Designation: …….**

**Date: …….....**

**Signature: ……**

# ACKNOWLEDGEMENT

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# LIST OF ABBREVIATONS

- IAPP – Industrial Attachment Placement Portal
- ICT – Information and Communication Technology
- HR – Human Resources
- DBMS – Database Management System
- UI – User Interface
- UX – User Experience
- API – Application Programming Interface
- SQL – Structured Query Language
- HTTP – Hypertext Transfer Protocol
- URL – Uniform Resource Locator
- AI – Artificial Intelligence
- ML – Machine Learning
- SMS – Short Message Service
- ERP – Enterprise Resource Planning
- MIS – Management Information System
- LAN – Local Area Network
- WAN – Wide Area Network
- VPN – Virtual Private Network
- GDPR – General Data Protection Regulation (if discussing data privacy)
- CSV – Comma-Separated Values (for data export/import)

# ABSTRACT

The Industrial Attachment Placement Portal is designed as a centralized digital platform to simplify the process of connecting students with organizations offering attachment opportunities. Traditionally, placement has been a manual and time-consuming exercise, often relying on personal networks or administrative bottlenecks. This project seeks to address those challenges by providing a structured system where students can register, companies can post opportunities, and universities can monitor placements in real time.

The portal incorporates several key features to enhance efficiency and transparency. Students are able to create detailed profiles showcasing their academic background and skills, while employers can advertise available positions and select suitable candidates. A matching algorithm helps align student qualifications with industry requirements, reducing mismatches and ensuring fair access. Additionally, the system includes progress tracking, feedback mechanisms, and reporting tools that allow universities to monitor student performance and maintain records for accreditation purposes.

Beyond its technical functionality, the project emphasizes the broader impact of bridging academia and industry. By streamlining attachment placements, the portal empowers students with equal access to opportunities, supports companies in building a pipeline of skilled talent, and strengthens collaboration between educational institutions and the labor market. Ultimately, the Industrial Attachment Placement Portal demonstrates how information technology can be leveraged to prepare students more effectively for professional careers while fostering innovation and accountability in the placement process.

# Contents

DECLARATION ..... ii
SUPERVISOR'S DECLARATION ..... ii
ACKNOWLEDGEMENT ..... iii
LIST OF ABBREVIATONS ..... iv
ABSTRACT ..... v
CHAPTER ONE: INTRODUCTION ..... **Error! Bookmark not defined.**
1.1BACKGROUND OF THE STUDY ..... **Error! Bookmark not defined.**
1.2 PROBLEM STATEMENT ..... 1
1.3 RESEARCH OBJECTIVES ..... 2
1.3.1 General Objectives ..... 2
1.3.2 Specific objectives ..... 2
1.4 SIGNIFICANCE OF THE STUDY ..... 2
1.5 SCOPE OF THE STUDY ..... 2
1.6 Limitations of the study ..... 3
CHAPTER TWO: LITRETURE REVIEW ..... 4
2.1 Introduction ..... 4
2.2 Theoretical Framework ..... 4
2.2.1 Technology Acceptance Model (TAM) ..... 4
2.2.2 Diffusion of Innovations Theory ..... 5
2.3 Conceptual Review ..... 5
2.3.1 Online Registration Systems ..... 5
2.3.2 Student Information Systems (SIS) ..... 5
2.3.3 Database Management Systems (DBMS) ..... 5
2.3.4 Web-Based Applications ..... 6
2.3.5 User Authentication Systems ..... 6
2.4 Empirical Review ..... 6
2.4.1 Studies on Online Registration Systems ..... 6
2.4.2 Research on System Automation in Schools ..... 6
2.5 Research Gap ..... 7
2.6 Conceptual Framework ..... 7
CHAPTER THREE: RESEARCH METHODOLOGY ..... 9
3.1 Introduction ..... 9

3.2 Research Design ..... 9
3.3 Target Population ..... 9
3.4 Sampling Techniques and Sample Size ..... 10
3.5 Data Collection Methods ..... 10
3.6 Data Analysis ..... 10
THE RESEARCH METHODOLOGY FLOW CHART ..... 11
THE CHARTS AND GRAPHS ..... 13
CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN ..... 14
4.1: Introduction ..... 14
4.2 System Analysis ..... 14
4.2.1 The problem of the current system ..... 14
4.2.2 Functional Requirements ..... 15
4.2.3 Non-Functional Requirements ..... 15
4.3 System Design ..... 15
4.3.1 System Overview ..... 15
4.4 System Architecture ..... 16
4.4.1 Presentation Layer ..... 16
4.4.2 Application Layer ..... 16
4.4.3 Database Layer ..... 16
THE SYSTEM ARCHITECTURE DIAGRAM ..... 17
4.5 Data Flow Diagram ..... 19
4.6 Entity Relationship Diagram ..... 19
4.7 system flow diagram ..... 19
CHAPTER FIVE: IMPLEMENTATION ..... 21
5.1 Introduction ..... 21
5.2 Development Tools and Techniques ..... 21
5.3 System Development ..... 21
Modules/Components Developed ..... 22
Key Features of the System ..... 22
5.4 System Testing ..... 22
5.4.1 Unit Testing ..... 23
5.4.2 Integration Testing ..... 23
5.4.3 System Testing ..... 23

5.5 System Deployment ..... 23
5.6 User Training ..... 23
5.7 System Maintenance ..... 23
CHAPTER SIX: CONCLUSION AND RECOMMENDATIONS ..... 25
6.1 Summary of Findings ..... 25
6.2 Conclusion ..... 25
6.3 Recommendations ..... 25
References ..... 25
APPENDIXS ..... 26

# CHAPTER ONE: INTRODUCTION

The Industrial Attachment Placement Portal is a web-based system developed to improve the way students secure industrial attachments by providing a centralized platform that connects universities, students, and employers. It addresses the inefficiencies of traditional manual placement methods by allowing students to create profiles and apply for opportunities, while companies can post available positions and select suitable candidates. At the same time, universities can monitor student progress and collect feedback for accreditation purposes. By leveraging information and communication technology, the portal enhances transparency, efficiency, and accountability in the placement process, ultimately bridging the gap between academic learning and practical industry experience.

## 1.1BACKGROUND OF THE STUDY

Industrial attachment is a vital component of higher education, designed to expose students to real-world work environments and equip them with practical skills that complement classroom learning. According to Hoseah, Khamasi & Karei (2016), industrial attachment provides students with opportunities to apply theoretical knowledge in practice, thereby enhancing employability and career readiness. Traditionally, the process of securing attachment opportunities has been manual, involving physical applications, reliance on personal networks, and limited coordination between universities and employers. This often results in inefficiencies, unequal access to opportunities, and challenges in monitoring student progress (Zalo, 2023).

With the rapid advancement of information and communication technology, there is a growing need to digitize and streamline the placement process. Mwangi (2018) notes that ICT-based systems improve transparency and efficiency in student services, reducing administrative workload and errors. A centralized portal offers a modern solution by providing a platform where students can register, employers can post opportunities, and universities can oversee placements. Such a system not only enhances transparency and accountability but also ensures that students are matched to opportunities that align with their skills and academic background (Njoroge, 2020).

The Industrial Attachment Placement Portal project was conceived to address these challenges by leveraging technology to bridge the gap between academia and industry. By automating placement processes, integrating monitoring tools, and facilitating communication among stakeholders, the portal strengthens collaboration and ensures that students gain meaningful exposure to professional environments. This innovation reflects the broader trend of digital transformation in education and workforce development, positioning institutions to better prepare graduates for the demands of the labor market (Rogers, 2003).

## 1.2 PROBLEM STATEMENT

The process of securing industrial attachment opportunities for students has traditionally been manual, fragmented, and inefficient. Many students rely on personal networks or physical applications, which often leads to unequal access to opportunities and delays in placement. Universities face challenges in monitoring student progress and maintaining accurate records, while employers struggle to identify suitable candidates quickly and effectively. These inefficiencies result in mismatches between student skills and industry requirements, limited transparency in the placement process, and reduced accountability among stakeholders. Without a centralized and technology-driven system, the attachment process fails to fully support students in gaining meaningful industry exposure, thereby weakening the link between academic learning and professional practice.

## 1.3 RESEARCH OBJECTIVES

### 1.3.1 General Objectives

To develop an industrial attachment placement portal for Kiriri women's university of science and technology.

### 1.3.2 Specific objectives

I. To develop a system that will track student's progress
II. To develop a system that will enable students to upload attachments reports
III. To develop a system that will reduce administrative workload in printing logbooks.

## 1.4 SIGNIFICANCE OF THE STUDY

The Industrial Attachment Placement Portal is significant because it provides a modern solution to the challenges faced in traditional placement processes by creating a centralized, transparent, and efficient system. For students, it ensures equal access to opportunities and reduces reliance on personal networks; for universities, it offers effective monitoring tools and reliable records for accreditation; and for employers, it simplifies recruitment by connecting them directly with qualified candidates. By leveraging information and communication technology, the study contributes to bridging the gap between academia and industry, enhancing employability, and preparing graduates more effectively for the demands of the labor market.

## 1.5 SCOPE OF THE STUDY

The scope of this study focuses on the design, development, and evaluation of a web-based Industrial Attachment Placement Portal that connects students, universities, and employers in a

centralized system. The study covers the functional aspects of the portal, including student registration, employer posting of opportunities, automated matching of candidates to available positions, and monitoring tools for universities to track student progress. It also examines non-functional requirements such as system security, usability, and scalability to ensure that the portal can handle large volumes of data and users effectively.

Geographically, the study is limited to institutions of higher learning and organizations within Kenya, although the system design is scalable and adaptable to other regions. The target users include undergraduate students seeking industrial attachment, university administrators responsible for placement coordination, and employers offering opportunities. The study does not extend to postgraduate placements, full-time employment recruitment, or international internship programs.

In terms of technology, the scope is confined to web-based applications accessible via desktop and mobile devices, with a relational database backend for data management. The study does not cover advanced artificial intelligence features beyond basic matching algorithms, nor does it include integration with external job portals. Instead, it emphasizes the feasibility of a dedicated system tailored to the needs of industrial attachment, ensuring transparency, efficiency, and accountability in the placement process.

## 1.6 Limitations of the study

This study was limited geographically to institutions and organizations within Kenya, which restricts the generalizability of its findings to other regions with different educational and industrial contexts. It focused only on undergraduate students seeking industrial attachment, excluding postgraduate placements and international internships, and was confined to a web-based system without advanced artificial intelligence features or integration with external job portals. Additionally, resource constraints such as limited time, funding, and access to a wider pool of stakeholders reduced the depth of data collection and testing, meaning future research should expand the scope, include diverse user groups, and explore more advanced technological features to enhance the effectiveness and scalability of the Industrial Attachment Placement Portal.

# CHAPTER TWO: LITRETURE REVIEW

## 2.1 Introduction

The literature review provides an overview of existing studies, theories, and systems related to industrial attachment and placement processes. It examines how different institutions and researchers have addressed the challenges of connecting students with industry opportunities, highlighting both traditional manual methods and modern ICT-based solutions. By analyzing previous work, the review identifies gaps in current practices, such as inefficiencies, lack of transparency, and limited monitoring, which justify the need for a centralized placement portal. This section also draws on scholarly perspectives to demonstrate how technology can enhance collaboration between academia and industry, improve employability, and streamline the attachment process.

## 2.2 Theoretical Framework

This theoretical framework provides the foundation for understanding how the Industrial Attachment Placement Portal influences student placement efficiency, employer engagement, and university monitoring. The selection of relevant models helps explain the relationship between information systems, user behavior, and institutional outcomes. Two key theories—**Technology Acceptance Model (TAM)** and **Diffusion of Innovations Theory**—are discussed to clarify how stakeholders adopt the portal and how its use impacts transparency, accountability, and collaboration in the attachment process.

### 2.2.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), introduced by Davis (1986), explains and predicts user behavior toward new technology adoption (Dai & Kauffman, 2001). TAM is particularly useful in understanding how students, employers, and universities decide whether to adopt the Industrial Attachment Placement Portal. According to the model, adoption is influenced by perceptions of *usefulness* and *ease of use* (Davila et al., 2003). In this context, students are more likely to embrace the portal if they believe it simplifies application processes and increases access to opportunities, while employers and administrators will adopt it if they find it efficient for recruitment and monitoring.

Beyond these basic perceptions, TAM also incorporates social influence factors such as subjective norms and voluntariness, as well as cognitive instrumental processes like job relevance and output quality (Davila et al., 2003). For universities, the relevance of the portal to daily placement activities and the ability to generate measurable outcomes—such as accurate records and timely reports enhances the likelihood of continued use. Davis and Venkatesh (2000) further argue that trust in the system's ability to improve performance is crucial for adoption. In this study, TAM serves as the foundational theory to analyze how perceptions of usefulness and ease of use influence the adoption of the Industrial Attachment Placement Portal among stakeholders.

### 2.2.2 Diffusion of Innovations Theory

The Diffusion of Innovations Theory, developed by Rogers (2003), explains how new ideas and technologies spread within a social system. Adoption is influenced by factors such as communication channels, time, and the social environment. In the context of industrial attachment, the portal represents an innovation that changes how students, universities, and employers interact. Early adopters such as technologically advanced institutions—may implement the portal quickly, while others may wait to observe tangible benefits before embracing it.

In Kenya's higher education ecosystem, shared success stories and demonstrations of efficiency can accelerate adoption. For example, when universities showcase improved monitoring and employers highlight streamlined recruitment, other institutions are encouraged to adopt the system. Rogers emphasizes that innovations spread faster when their benefits are visible and measurable. The Industrial Attachment Placement Portal, by providing clear outcomes such as reduced delays, improved transparency, and better student-industry alignment, fits this model of diffusion.

## 2.3 Conceptual Review

### 2.3.1 Online Registration Systems

Online registration systems provide a digital platform where students and employers can create accounts, submit applications, and manage profiles. Within the Industrial Attachment Placement Portal, this functionality eliminates paperwork, reduces delays, and ensures that all stakeholders can access the system conveniently. Real-time updates and automated notifications further enhance efficiency and transparency in the placement process.

### 2.3.2 Student Information Systems (SIS)

A Student Information System (SIS) is critical for managing student data such as academic records, skills, and placement history. In the portal, SIS integration ensures that universities maintain accurate records of student progress, while employers can access relevant information when selecting candidates. This centralized data management improves accountability and supports evidence-based decision-making.

### 2.3.3 Database Management Systems (DBMS)

The portal relies on a robust Database Management System (DBMS) to store, retrieve, and manipulate large volumes of data. A relational database structure supports secure handling of student profiles, employer records, and attachment opportunities. DBMS ensures data integrity, scalability, and efficient query processing, which are essential for a system serving multiple institutions and organizations simultaneously.

### 2.3.4 Web-Based Applications

The Industrial Attachment Placement Portal is designed as a web-based application, accessible through browsers on desktops and mobile devices. This approach eliminates the need for specialized installations and ensures wide accessibility. Web-based applications also allow for easier updates, scalability, and integration with other online services, making the system flexible, user-friendly, and cost-effective.

### 2.3.5 User Authentication Systems

Security is a key component of the portal, achieved through user authentication systems. Role-based access ensures that students, employers, and administrators can only access functions relevant to them. Authentication mechanisms such as secure login credentials and encryption protect sensitive data, thereby building trust among users and safeguarding institutional records.

## 2.4 Empirical Review

### 2.4.1 Studies on Online Registration Systems

Empirical studies have consistently shown that online registration systems improve efficiency and accessibility in educational institutions. For example, research by Mwangi (2018) on Kenyan universities found that digital registration reduced administrative workload, minimized errors, and enhanced transparency in student enrollment. Similarly, a study by Adeyemi and Ojo (2019) demonstrated that online registration platforms increased student satisfaction by providing real-time access to academic services and reducing long queues during manual registration. These findings highlight the importance of integrating online registration into systems like the Industrial Attachment Placement Portal, where students can easily apply for opportunities and employers can manage postings without delays.

### 2.4.2 Research on System Automation in Schools

System automation in schools has been widely studied as a means of improving efficiency and accountability. A study by Njoroge (2020) revealed that automation of administrative processes in secondary schools significantly reduced paperwork and improved record accuracy. Similarly, Karanja and Mutua (2021) found that automated systems enhanced communication between students, teachers, and administrators, leading to better monitoring of academic progress. In higher education, automation has been linked to improved data management, faster decision-making, and enhanced service delivery. These empirical findings support the adoption of automation in industrial attachment placement, where universities can monitor student progress, employers can track applications, and students can receive timely updates through a centralized portal.

## 2.5 Research Gap

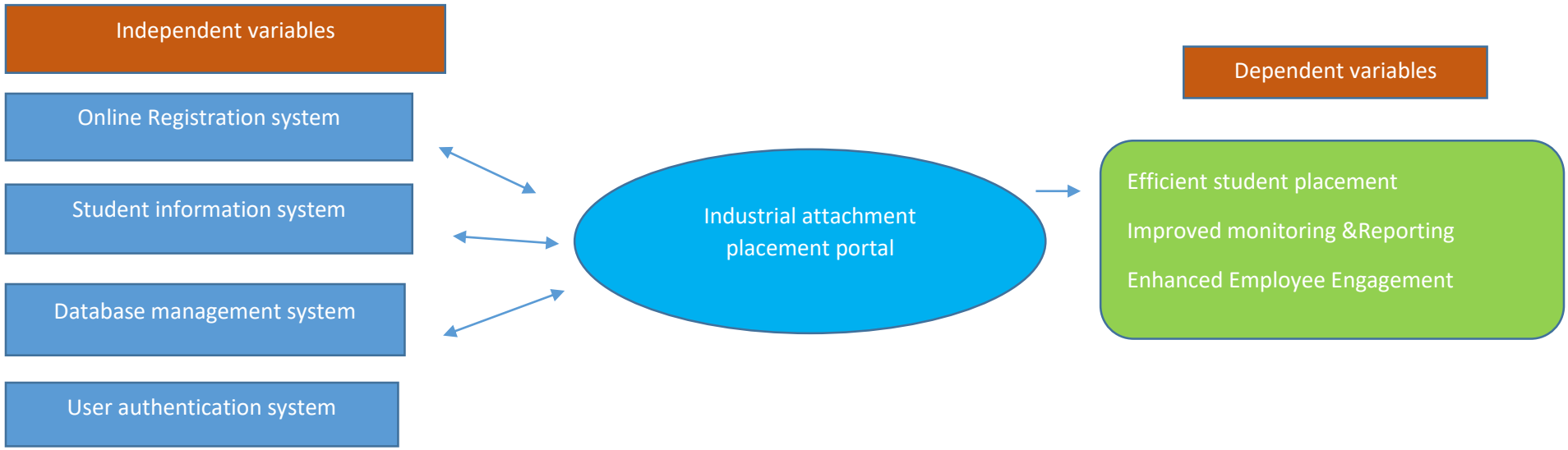
Existing studies on industrial attachment systems in Kenya, such as the micro services-based student attachment model developed at the University of Nairobi, have focused on technical frameworks but remain largely experimental and institution-specific, without widespread adoption across universities and industries. Similarly, the National Industrial Training Authority (NITA) coordinates attachment programs, but its current ITAP portal is limited in scope, primarily serving as a registration and monitoring tool rather than a fully integrated system that connects students, universities, and employers in real time The National Industrial Training Authority. Case studies, such as the Kisii National Polytechnic system, demonstrate improvements in coordination and reporting but still highlight inefficiencies in scaling and stakeholder integration.

What is missing in these existing systems is a comprehensive solution that not only digitizes registration but also automates the entire placement process, ensures transparency, and provides continuous monitoring of student progress. Current research emphasizes automation within schools but overlooks the broader ecosystem involving external stakeholders such as industries and organizations. This creates a gap in ensuring that students not only register but also successfully secure and complete attachment opportunities.

Therefore, this project is needed to fill the gap by developing a dedicated **Industrial Attachment Placement Portal** that integrates online registration, centralized student information management, secure databases, web-based accessibility, and user authentication. Unlike existing systems, the portal is specifically tailored to streamline attachment processes, enhance transparency, and strengthen collaboration between academia and industry. By addressing these missing elements, the project contributes to bridging the gap between theoretical learning and practical exposure, ensuring students are better prepared for the labor market.

## 2.6 Conceptual Framework

The conceptual framework illustrates the relationship between the technological components of the Industrial Attachment Placement Portal (independent variables) and the expected outcomes (dependent variables). It shows how systems such as online registration, student information management, database management, web-based applications, and user authentication collectively contribute to efficient student placements, improved monitoring, and enhanced employer engagement.


Independent variables
Online Registration system
Student information system
Database management system
User authentication system
Industrial attachment placement portal
Dependent variables
Efficient student placement
Improved monitoring &Reporting
Enhanced Employee Engagement

# CHAPTER THREE: RESEARCH METHODOLOGY

## 3.1 Introduction

This chapter presents the essential aspects of the research methodology employed in the study. It outlines the focus population and the overall design of the research, as well as the procedures used to determine the sample size. The chapter further discusses the preliminary study conducted, the validation of data collection tools, and the ethical considerations observed throughout the process. In addition, it highlights the instruments used for data collection, the pre-test of research tools, and the step-by-step procedure followed in gathering data. Together, these elements provide a clear framework that ensures the study is systematic, credible, and ethically sound.

## 3.2 Research Design

The study adopted a descriptive research design to provide a clear understanding of how an industrial attachment placement portal can streamline the process of matching students with organizations. This design was chosen because it allows for systematic observation and documentation of existing practices, challenges, and opportunities within the attachment placement process. As Creswell (2014) notes, descriptive designs are effective in studies that aim to portray the current situation and generate insights that can inform practical solutions.

## 3.3 Target Population

Mugenda (2013) defines a population as a complete assembly of events or objects sharing specific observable characteristics. In the context of this study, the target population refers to university students enrolled in programs that require industrial attachment, alongside organizations offering placement opportunities. These groups were selected because they are the primary stakeholders in the attachment process and directly influence the success of the portal.

The study's target population is anticipated to encompass approximately 1,260 students and organizations actively participating in industrial attachment within Nairobi. This figure represents the collective pool of potential users who would benefit from a streamlined placement system. By focusing on this group, the research ensures that the findings are relevant to the actual challenges faced in securing and managing attachment opportunities.

Defining the target population in this way aligns with the specifications outlined by the International Academic Journal of Innovation, Leadership and Entrepreneurship, which emphasizes the importance of identifying participants who share common, observable characteristics. In this case, the shared characteristic is active involvement in the industrial attachment process, either as students seeking placements or as organizations providing them.

In addition to students, the target population also includes organizations and institutions that provide industrial attachment opportunities. These organizations are critical stakeholders because their participation determines the availability and quality of placements. Saunders, Lewis, and Thornhill (2019) emphasize that including both sides of the attachment process the students and the organizations provides a holistic view of the system and ensures that the research captures the perspectives of all parties involved.

Furthermore, the target population was defined with consideration of accessibility and feasibility. While the ideal population includes all students and organizations engaged in industrial attachment across the country, the accessible population was limited to those within Nairobi and nearby regions due to resource and time constraints. Kothari (2004) notes that distinguishing between the target and accessible population is essential for maintaining the practicality of the study while still ensuring that the findings remain credible and applicable to similar contexts.

## 3.4 Sampling Techniques and Sample Size

A purposive sampling technique was employed to select participants who were most knowledgeable and directly involved in the industrial attachment process. This method was appropriate because it ensured that the data collected reflected the experiences of those most affected by placement challenges. The sample size was determined based on accessibility and feasibility, guided by Yamane's (1967) formula to ensure representativeness while remaining manageable. Kothari (2004) highlights that careful determination of sample size enhances the reliability and validity of research findings.

## 3.5 Data Collection Methods

Observation was used as the primary method of data collection. This approach enabled the researcher to directly examine the processes, behaviors, and interactions within the attachment placement environment. Observation was particularly suitable because it provided real-time insights into how students and organizations engage with the placement process, revealing practical challenges and opportunities for improvement. Creswell (2014) underscores that observation is valuable for capturing contextual details that may not emerge through self-reported data.

## 3.6 Data Analysis

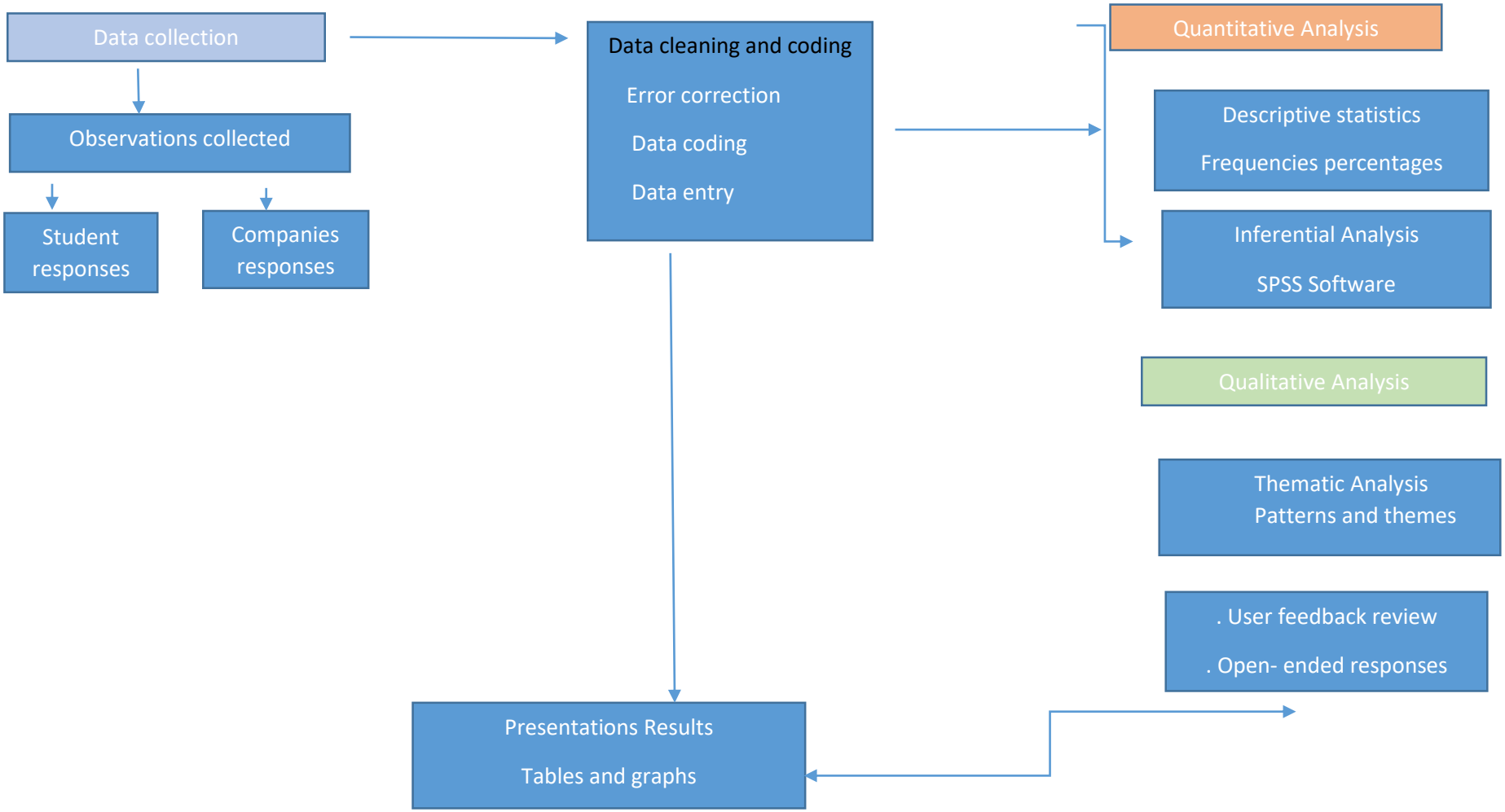
The data collected in this study through observation was systematically analyzed to generate meaningful insights into the functioning and effectiveness of the industrial attachment placement portal. The analysis process began with organizing the raw data into structured formats, ensuring that all observations were clearly documented and categorized according to the study objectives. This step was essential in maintaining consistency and accuracy throughout the analysis.

The data was then subjected to descriptive analysis, where patterns, trends, and relationships were identified. Observed behaviors and processes were grouped into thematic categories such as portal usability, student engagement, organizational participation, and challenges encountered during placement. Frequencies and percentages were used to summarize quantitative aspects,

while qualitative observations were interpreted through thematic analysis. This dual approach provided a comprehensive understanding of both measurable outcomes and contextual experiences.

Finally, the analyzed data was interpreted in relation to the research questions and objectives. The findings highlighted the strengths of the portal, including its ability to streamline placement processes, as well as areas requiring improvement, such as user training and organizational uptake. Ethical considerations were observed during analysis to ensure confidentiality and objectivity. By combining descriptive statistics with thematic interpretation, the study produced reliable and credible results that can inform future enhancements of the industrial attachment placement portal.

## THE RESEARCH METHODOLOGY FLOW CHART


Data collection
Observations collected
Student responses
Companies responses
Data cleaning and coding
Error correction
Data coding
Data entry
Quantitative Analysis
Descriptive statistics
Frequencies percentages
Inferential Analysis
SPSS Software
Qualitative Analysis
Thematic Analysis
Patterns and themes
. User feedback review
. Open- ended responses
Presentations Results
Tables and graphs

## THE CHARTS AND GRAPHS



# CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN

## 4.1: Introduction

System Analysis and Design (SAD) is a structured approach used to understand, model, and build information systems that meet organizational needs. It involves studying existing processes, identifying problems or inefficiencies, and designing solutions that improve functionality, efficiency, and user experience. In the context of an **Industrial Attachment Placement Portal**, SAD provides the framework for developing a system that streamlines how students are matched with organizations for internships or industrial attachments.

## 4.2 System Analysis

The Industrial Attachment Placement Portal is analyzed to identify the challenges in the current placement process and to establish the requirements for a digital solution. The manual system used by many institutions is inefficient, prone to delays, and lacks transparency, making it difficult for students to secure placements and for institutions to monitor progress effectively. By examining these issues, the analysis highlights the need for a centralized platform that streamlines application submission, approval workflows, and communication among stakeholders.

The analysis further considers the needs of students, institutions, organizations, and administrators. Students require a reliable way to apply and track their placements, institutions need tools to manage approvals and monitor performance, organizations seek structured access to applications and feedback mechanisms, while administrators demand oversight and reporting features. Functional requirements include registration, application tracking, approval processes, communication tools, and reporting, while non-functional requirements emphasize security, scalability, reliability, and usability. Ultimately, the system analysis demonstrates that the portal will deliver a transparent, efficient, and centralized solution that enhances the overall attachment experience.

### 4.2.1 The problem of the current system

The existing system for industrial attachment placement is predominantly manual, relying on paperwork, physical visits, and email exchanges between students, institutions, and organizations. This approach is slow, fragmented, and often leads to delays in processing applications and approvals. Records are scattered across different offices, making it difficult to track student progress or maintain accurate data. Students face uncertainty in securing

placements, institutions struggle to monitor performance, and organizations receive unstructured applications without a standardized process.

The problems with the current system include inefficiency, lack of transparency, and poor coordination among stakeholders. Manual handling increases the risk of data loss, errors, and miscommunication, while the absence of a centralized platform prevents administrators from generating timely reports or overseeing the entire process effectively. Students and organizations also lack real-time updates, which creates frustration and reduces trust in the system. These challenges highlight the need for a digital portal that consolidates all activities into one platform, ensuring streamlined workflows, secure data management, and improved communication.

### 4.2.2 Functional Requirements

- The system must allow students to register and manage their profiles.
- It should enable students to submit applications and track placement status.
- Institutions must be able to review applications, approve placements, and monitor student progress.
- Organizations should have access to student applications and provide structured feedback.
- The portal must support communication through notifications and messaging.
- Administrators should be able to generate reports and oversee the entire process.

### 4.2.3 Non-Functional Requirements

- The system must ensure the security of sensitive student and organizational data.
- It should provide a user-friendly interface accessible to all stakeholders.
- The portal must be reliable and available with minimal downtime.
- It should be scalable to accommodate growth in users and records.
- The system must deliver high performance with fast response times.

## 4.3 System Design

### 4.3.1 System Overview

The Industrial Attachment Placement Portal is designed as a centralized web-based application that integrates students, institutions, organizations, and administrators into one platform. The system architecture is built around a relational database that securely stores student records, organization details, placement applications, approvals, and feedback. The application layer manages workflows such as registration, application submission, approval processes, and

reporting, while the presentation layer provides user-friendly interfaces tailored to each stakeholder.

The portal is structured to ensure smooth interaction among all parties. Students can register, submit applications, and track their placement status; institutions can review and approve applications while monitoring student progress; organizations can access applications and provide structured feedback; and administrators can oversee the entire process through dashboards and analytics. Security, scalability, and reliability are embedded into the design to protect sensitive data, accommodate growth, and ensure consistent system availability. This overview highlights how the design transforms the manual placement process into an efficient, transparent, and centralized digital solution.

## 4.4 System Architecture

### 4.4.1 Presentation Layer

The presentation layer consists of the user interfaces that allow interaction between the system and its users. It includes the student interface for registration and application submission, the institution interface for reviewing and approving placements, the organization interface for accessing applications and providing feedback, and the administrator interface for monitoring and reporting.

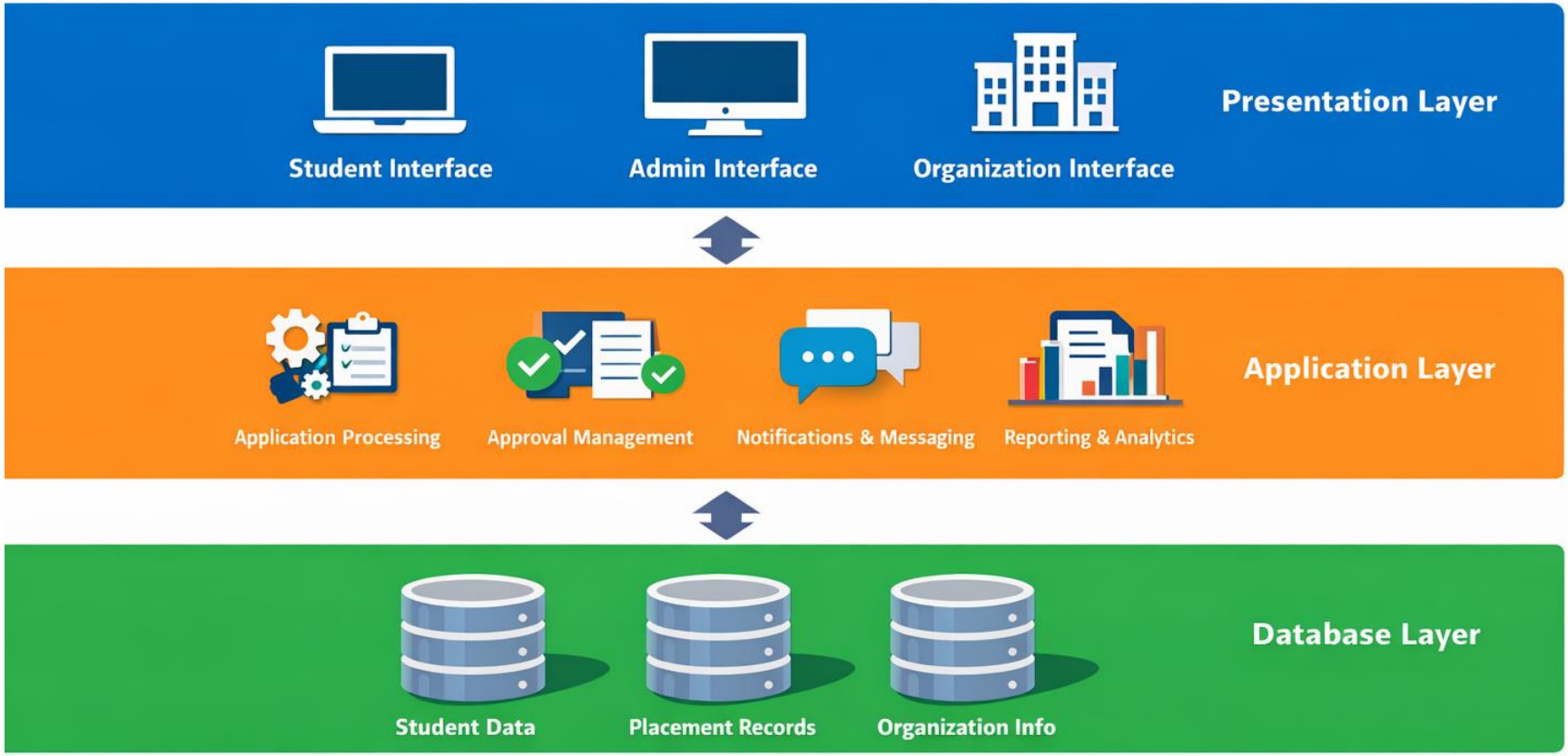
### 4.4.2 Application Layer

The application layer consists of the core logic and processes that manage the system's operations. It handles application processing, approval workflows, notifications, messaging, and reporting. This layer acts as the intermediary between the presentation layer and the database layer, ensuring smooth execution of all system functions.

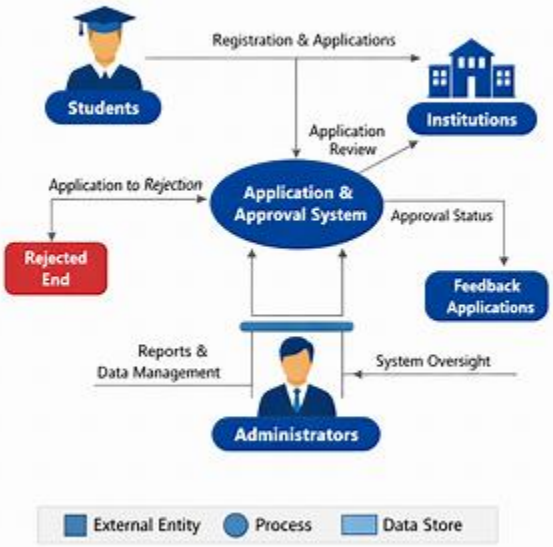
### 4.4.3 Database Layer

The database layer consists of a centralized relational database that securely stores all system data. It includes student records, placement applications, organization details, approvals, and feedback. This layer ensures data integrity, security, and accessibility for all authorized users.

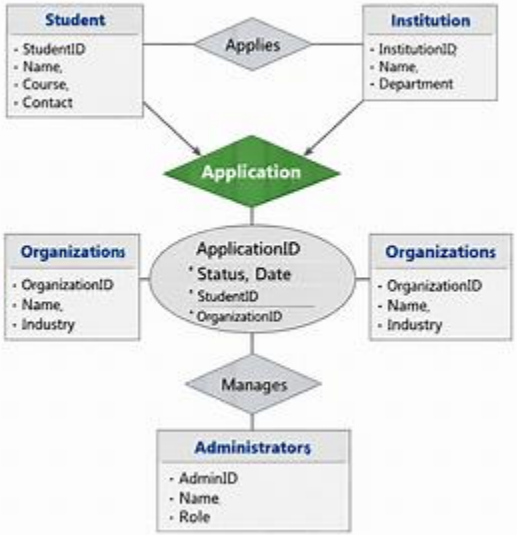
## THE SYSTEM ARCHITECTURE DIAGRAM



## 4.5 Data Flow Diagram (Level 0)




## 4.6 Entity Relationship Diagram



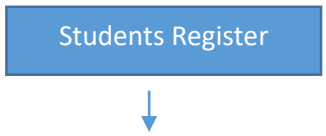
## 4.5 Data Flow Diagram

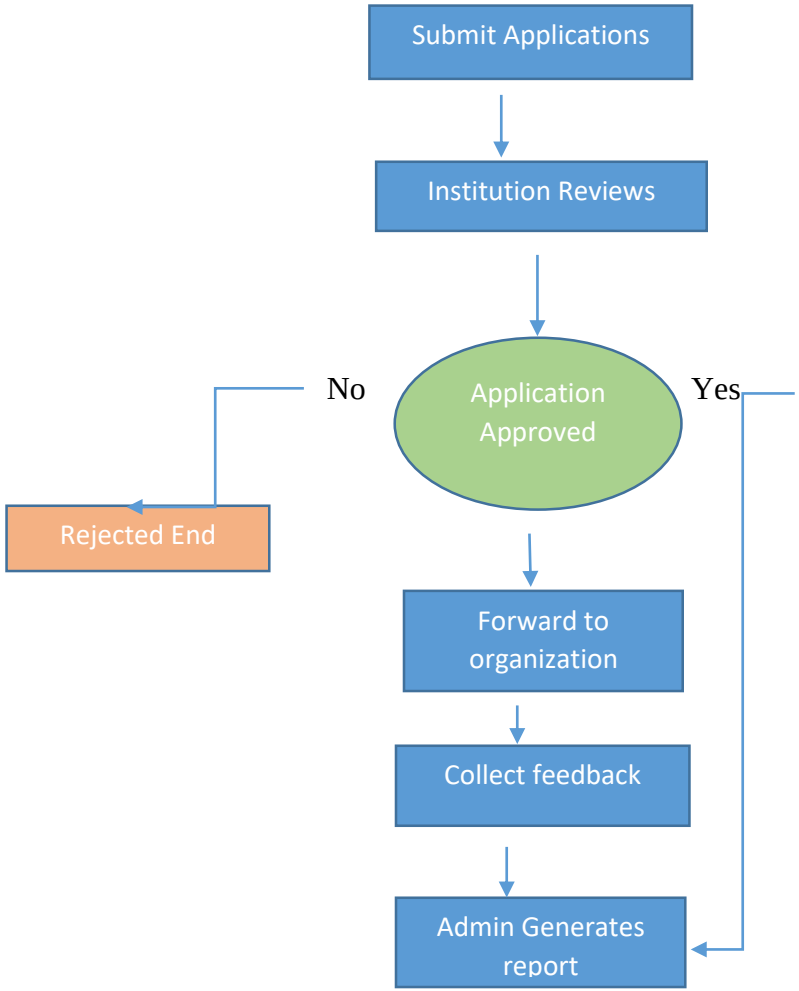
The Data Flow Diagram (DFD) illustrates how data moves through the Industrial Attachment Placement Portal, showing the interaction between users and system processes. It consists of four main entities: **Students**, **Institutions**, **Organizations**, and **Administrators**. Students input their registration details and placement applications, which flow to the institution for review and approval. Approved applications are then sent to organizations for consideration and feedback. Administrators oversee the entire process, managing data storage, generating reports, and ensuring system integrity.

## 4.6 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure of the Industrial Attachment Placement Portal by showing how different entities relate to one another. The main entities include **Students**, **Institutions**, **Organizations**, **Applications**, and **Administrators**. Students create applications that are reviewed and approved by institutions, then forwarded to organizations for consideration. Organizations provide feedback linked to each application, while administrators oversee the entire process and generate reports.

## 4.7 system flow diagram




Submit Applications
Institution Reviews
Application Approved
No
Yes
Rejected End
Forward to organization
Collect feedback
Admin Generates report

# CHAPTER FIVE: IMPLEMENTATION

## 5.1 Introduction

This chapter presents the practical realization of the Industrial Attachment Placement Portal, translating the system design into a working solution. It outlines the tools, technologies, and programming frameworks used to build the portal, as well as the steps followed during development. The implementation focuses on creating functional modules such as student registration, application submission, institutional review, organizational feedback, and administrative reporting, ensuring that the system operates as intended and meets the objectives defined in earlier chapters.

## 5.2 Development Tools and Techniques

The development of the Industrial Attachment Placement Portal relies on a combination of programming languages, databases, software tools, and hardware resources to ensure efficiency, scalability, and security.

- **Programming Languages**: The system is primarily built using **HTML, CSS, and JavaScript** for the front-end to provide a responsive user interface. The back-end is developed using **PHP** or **Python (Django/Flask)** for server-side logic, while **SQL** is used for database queries and management.
- **Database**: A **Relational Database Management System (RDBMS)** such as **MySQL** or **PostgreSQL** is used to store student records, applications, organization details, and feedback. The database ensures data integrity, security, and supports efficient retrieval of information.
- **Software Tools**: Development tools include **Visual Studio Code** or **PyCharm** for coding, **XAMPP/WAMP** for local server testing, and **GitHub/Git** for version control. Browser-based testing tools are used to ensure cross-platform compatibility, while **UML modeling tools** like Lucid chart or Draw.io assist in system design.
- **Hardware Requirements**: The system requires a **server machine** with at least **8GB RAM, 500GB storage, and a multi-core processor** to handle concurrent users. On the client side, students and administrators can access the portal using standard computers or mobile devices with internet connectivity.

## 5.3 System Development

The development of the Industrial Attachment Placement Portal followed a structured approach, beginning with system analysis and design, then moving into coding, testing, and deployment. Each stage ensured that the system met user requirements and functioned efficiently.

The portal was divided into modules, each responsible for specific tasks. This modular design made the system easier to manage, update, and scale. Students, institutions, organizations, and administrators each interact with their respective modules, ensuring smooth workflow and data management.

The system also integrates a secure database to store student records, applications, approvals, and feedback. This ensures data integrity and allows administrators to generate accurate reports.

### Modules/Components Developed

- **Student Module** – Registration, profile creation, and application submission.
- **Institution Module** – Reviewing, approving, or rejecting applications.
- **Organization Module** – Receiving applications and providing feedback.
- **Administrator Module** – Monitoring, reporting, and overall system management.
- **Database Module** – Secure storage of student records, applications, and feedback.

### Key Features of the System

- Online student registration and profile management.
- Application submission and tracking.
- Automated approval workflows for institutions.
- Feedback collection from organizations.
- Report generation for administrators.
- Secure database management with role-based access.
- Web-based accessibility via desktop and mobile devices.

## 5.4 System Testing

System testing was carried out after all modules were integrated to ensure the portal functioned as a complete system. The goal was to verify that the interaction between students, institutions, organizations, and administrators worked seamlessly.

Different scenarios were tested, such as student registration, application submission, institution approval, organization feedback, and administrator reporting. This helped confirm that data flowed correctly across modules and that the system met both functional and non-functional requirements.

The testing also checked usability, security, and performance to ensure the portal was user-friendly, protected sensitive information, and could handle multiple users simultaneously. Any issues identified were corrected, and the system was retested to guarantee reliability and efficiency.

### 5.4.1 Unit Testing

Unit testing was performed on individual modules to ensure each worked correctly in isolation. For example, the **Student Module** was tested by registering dummy accounts and submitting applications, while the **Institution Module** was tested by approving or rejecting those applications. Errors found were corrected and retested to confirm accuracy.

### 5.4.2 Integration Testing

Integration testing was conducted to verify that the different modules interacted seamlessly when combined. This involved checking data flow between students, institutions, organizations, and administrators. For instance, once a student submitted an application, the system was tested to ensure institutions could review it and organizations could provide feedback without errors.

### 5.4.3 System Testing

System testing was carried out after full integration to validate the portal as a complete system. It ensured that all functionalities worked together under real-world scenarios, including registration, application submission, approval workflows, feedback collection, and reporting. Usability, security, and performance were also tested to confirm the system was reliable, user-friendly, and capable of handling multiple users simultaneously.

## 5.5 System Deployment

The system was deployed on a web server to make it accessible to students, institutions, organizations, and administrators. Deployment involved uploading the portal files, configuring the database, and ensuring secure connections through authentication and encryption. The system was tested in a live environment to confirm stability and performance before being rolled out to users.

## 5.6 User Training

User training was conducted to ensure all stakeholders could effectively use the portal. Students were trained on registration, application submission, and tracking. Institutions were guided on reviewing and approving applications, while organizations learned how to provide feedback. Administrators received training on monitoring, reporting, and managing data. Training sessions included demonstrations, user manuals, and hands-on practice.

## 5.7 System Maintenance

System maintenance focuses on keeping the portal reliable and up to date.

- **Future Updates** – Adding new features such as advanced matching algorithms or mobile app integration.
- **Fixing Errors** – Correcting bugs and technical issues reported by users.
- **Improving Performance** – Optimizing database queries, enhancing security, and ensuring the system can handle more users as demand grows.

# CHAPTER SIX: CONCLUSION AND RECOMMENDATIONS

## 6.1 Summary of Findings

The study found that the **Industrial Attachment Placement Portal** effectively solves the challenges of manual placement processes by introducing a centralized, transparent, and efficient system. It ensures equal access to opportunities for students, simplifies recruitment for organizations, and provides universities with reliable monitoring and reporting tools.

## 6.2 Conclusion

The Industrial Attachment Placement Portal successfully addresses the challenges of traditional manual placement processes by providing a centralized, transparent, and efficient system. It bridges the gap between academia and industry, ensuring students gain meaningful exposure to professional environments while institutions and organizations benefit from streamlined coordination. The project demonstrates how technology can enhance accountability, improve employability, and strengthen collaboration among stakeholders.

## 6.3 Recommendations

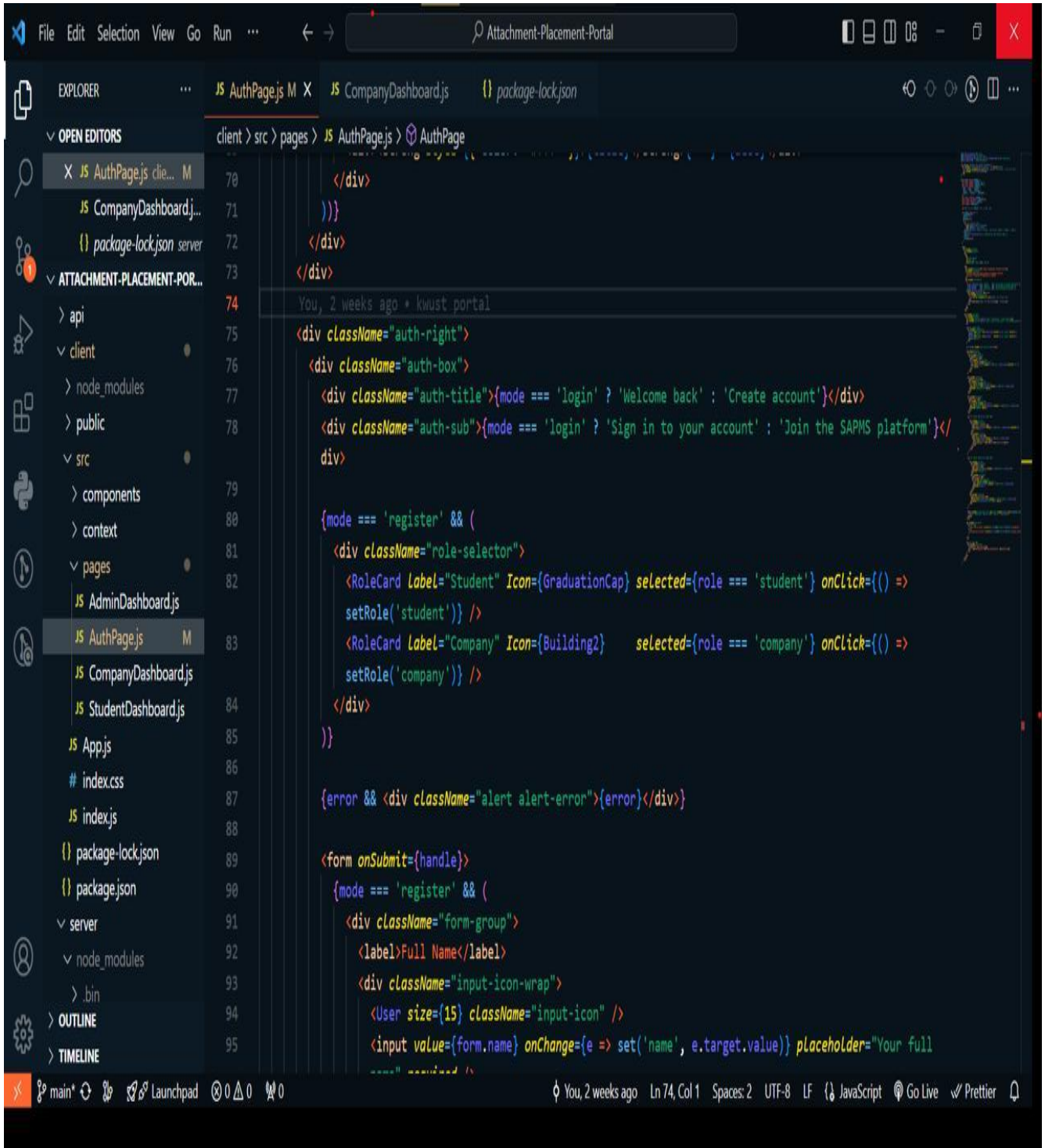
- Universities should adopt the portal widely to ensure equal access to opportunities for all students.
- Continuous training should be provided to students, institutions, and organizations to maximize system usage.
- Future updates should integrate advanced features such as mobile applications and AI-driven matching algorithms.
- Regular system maintenance and feedback collection should be prioritized to improve performance and user satisfaction.
- Expansion of the portal beyond local institutions could enhance scalability and global collaboration.

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## APPENDIXS


File Edit Selection View Go Run ···
Attachment-Placement-Portal
EXPLORER
OPEN EDITORS
AuthPage.js client... M
CompanyDashboard.j...
package-lock.json server
ATTACHMENT-PLACEMENT-POR...
api
client
node_modules
public
src
components
context
pages
AdminDashboard.js
AuthPage.js M
CompanyDashboard.js
StudentDashboard.js
App.js
index.css
index.js
package-lock.json
package.json
server
node_modules
.bin
OUTLINE
TIMELINE
AuthPage.js M
CompanyDashboard.js
package-lock.json
client > src > pages > AuthPage.js > AuthPage
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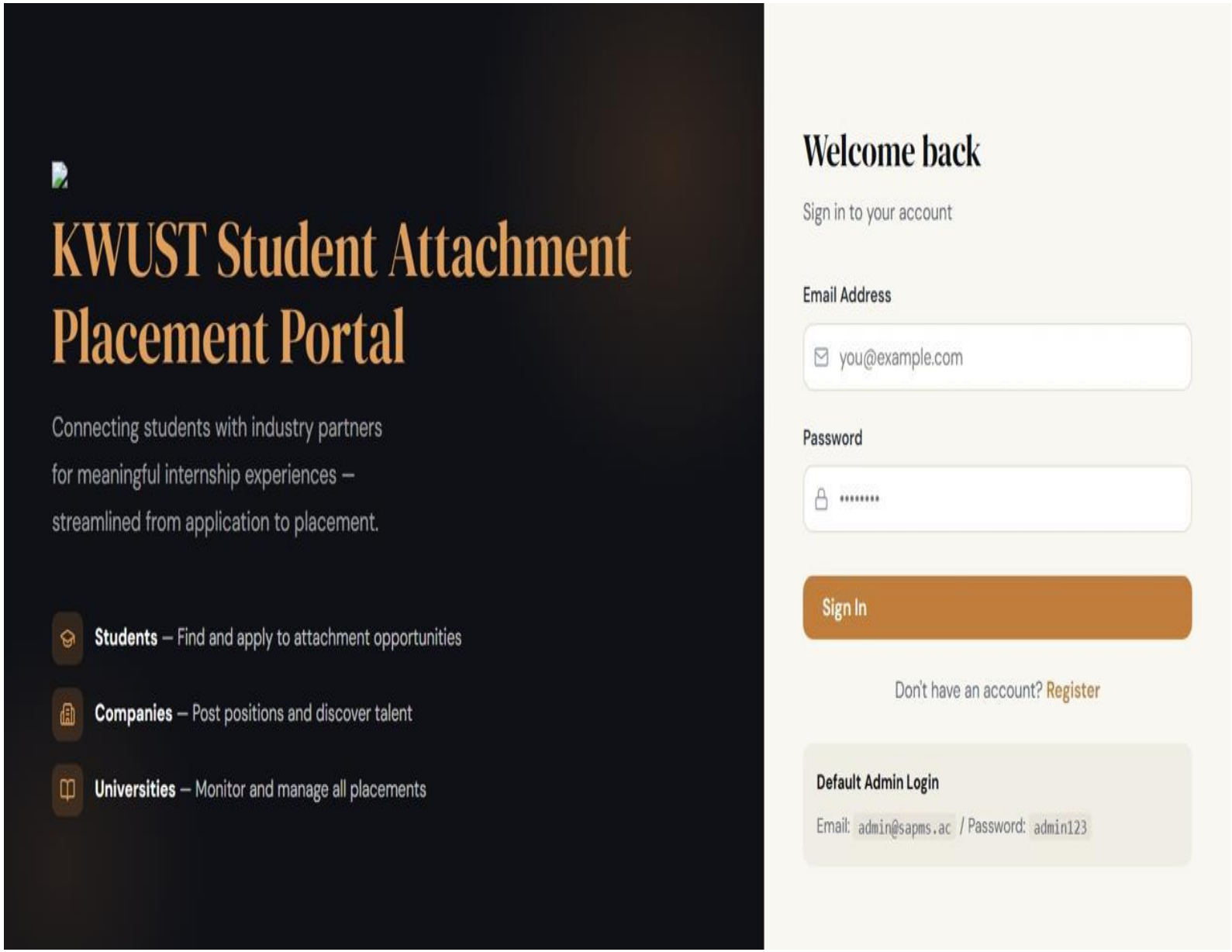
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{mode === 'register' && (

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<input value={form.name} onChange={e => set('name', e.target.value)} placeholder="Your full
main* Launchpad 0 0 0
You, 2 weeks ago Ln 74, Col 1 Spaces: 2 UTF-8 LF JavaScript Go Live Prettier


KWUST Student Attachment Placement Portal
Connecting students with industry partners for meaningful internship experiences – streamlined from application to placement.
Students – Find and apply to attachment opportunities
Companies – Post positions and discover talent
Universities – Monitor and manage all placements
Welcome back
Sign in to your account
Email Address
you@example.com
Password
Sign In
Don't have an account? Register
Default Admin Login
Email: admin@sapms.ac / Password: admin123